PROBLEM NS10

Underwater Explosion Cavity

Original statement · NS10.pdf

Spherical symmetry | Null stress condition | Non-dimensionalization | Limits Re large and small

OFFICIAL PROBLEM STATEMENT

As a consequence of an underwater explosion, a spherical cavity of initial radius R_0 and initial radial velocity V_0 is created inside the ocean. The growth of the cavity induces a flow in the surrounding liquid, whose density ρ and viscosity μ can be considered constant. It is also assumed that the effect of gravity is negligible, and that the motion is sphericosymmetric, $\bar{v} = v_r(r,t)\,\bar{e}_r$, where $R_c(t) \leq r < \infty$, being $R_c(t)$ the radius of the cavity at time t. The liquid is at rest far from the cavity, $r \to \infty$, where the pressure is p_∞, while the pressure inside the cavity can be neglected compared to p_∞.

1. (1.5 points). Find the radial velocity of the liquid, $v_r(r,t)$, as a function of the radius of the cavity, $R_c(t)$, and of its associated expansion velocity, $\dot{R}_c(t) = dR_c/dt$. Show that the resulting flow is irrotational.

2. (1.5 points). Obtain the pressure field, $p(r,t)$, as a function of $R_c(t)$, $\dot{R}_c(t)$, $\ddot{R}_c(t)$, ρ and p_∞.

3. (1 point). Evaluate the resulting pressure distribution at $r = R_c(t)$, to find the ordinary differential equation (ODE) with initial conditions satisfied by $R_c(t)$, in terms of the unknown liquid pressure at the cavity boundary, $p_R(t)$.

4. (1.5 points). Find $p_R(t)$ by imposing the condition of null stress of the liquid at the cavity wall, $\bar{\bar{\tau}} \cdot \bar{n}\big|_{r=R_c(t)} = 0$, where $\bar{\bar{\tau}} = -p\bar{\bar{I}} + \bar{\bar{\tau}}'$ is the stress tensor.

5. (1.5 points). Using the answer to question #4, write the ODE obtained in question #3 in closed form. Nondimensionalize the corresponding initial value problem, using the following dimensionless variables: $R = R_c/R_0$, $\tau = tV_0/R_0$. Show that the problem is governed by two dimensionless parameters, namely the Reynolds number, $\mathrm{Re} = \rho V_0 R_0/\mu$, and $\Lambda = p_\infty/(\rho V_0^2)$.

6. (1.5 points). Simplify the ODE in the limit $\mathrm{Re} \to \infty$. By using $z = \ln(R)$ as the independent variable and $F = (dR/d\tau)^2$ as the dependent one, show that $F(z)$ satisfies the linear ODE

$$\frac{dF}{dz} + 3F = -2\Lambda,$$

with initial condition $F(0) = 1$. Obtain $F(z)$, and use it to find $dR/d\tau$ as a function of R.

7. (1.5 points). Obtain the maximum radius of the cavity, $R_{\max}$, as a function of Λ, and show that the time $\tau_{\max}$, defined such that $R(\tau_{\max}) = R_{\max}$ is given by the integral

$$\tau_{\max}(\Lambda) = \int_1^{(1+\frac{3}{2\Lambda})^{1/3}} \frac{dx}{\sqrt{x^{-3} + \frac{2\Lambda}{3}\,(x^{-3}-1)}}.$$

Provide an explicit approximation for $\tau_{\max}$ in the limit $\Lambda \ll 1$.

8. (1 additional point). Simplify the ODE found in question #5 in the limit $\mathrm{Re} \to 0$. Is it possible to satisfy the initial conditions? Find an approximate solution, and discuss its validity.

The sheet is headed "INTRODUCTION TO FLUID MECHANICS, Dec 18, 2012, PROBLEM 2 (1 hour)" and carries no figure; the spelling "sphericosymmetric" is the sheet's.

`Source: NS10.pdf (page 1 of 3: statement; pages 2 and 3: handwritten solution). Notes.pdf, Chapter 4: continuity Eq. 4.40, momentum Eq. 4.41 in spherical coordinates, stress components in spherical coordinates.`

Worked solution (4-phase method)

`PHASE 1`

Hypotheses, Data and Unknowns

- Incompressible liquid with constant ρ and μ, spherically symmetric flow $\bar{v} = v_r(r,t)\,\bar{e}_r$ in $r > R_c(t)$, gravity neglected. Equations (4.40) and (4.41) in spherical coordinates.
- The cavity gas pressure is neglected compared with p_∞: the liquid has no traction applied at the cavity wall (null stress). The liquid is at rest and at p_∞ far away.
- Initial conditions: $R_c(0) = R_0$, $\dot{R}_c(0) = V_0$.
- Data: $[\rho,\ \mu,\ p_\infty,\ R_0,\ V_0]$. Unknowns: $v_r, p, p_R(t)$, the ODE for $R_c(t)$ and its non-dimensional limits.

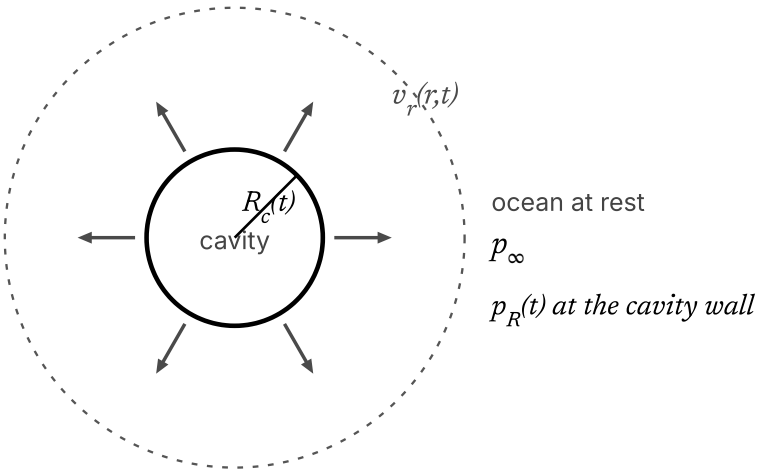
Symbol	Meaning	SI unit
$R_c(t)$, $\dot{R}_c$, $\ddot{R}_c$	cavity radius, velocity and acceleration	m, m/s, m/s^2
R_0, V_0	initial radius and initial expansion velocity	m, m/s
$p_R(t)$	liquid pressure at the cavity wall	Pa
$\mathrm{Re} = \rho V_0 R_0/\mu$, $\Lambda = p_\infty/(\rho V_0^2)$	dimensionless parameters	1

Symbol	Meaning	SI unit
$R = R_c/R_0, \tau = tV_0/R_0$	dimensionless radius and time	1

PHASE 2

Frames, Field Form and Boundary Conditions

- Spherical frame (r, θ, ϕ) centred on the cavity centre, fixed; the liquid occupies $r > R_c(t)$. The basis is orthonormal and right-handed.
- Field form: $\bar{v} = v_r(r,t)\,\bar{e}_r, p = p(r,t)$.



Setup of NS10 (drawn for the solution): cavity of radius $R_c(t)$ ($R_c(0) = R_0$, $\dot{R}_c(0) = V_0$) expanding in the ocean, radial velocity field $v_r(r,t)$, pressure $p_R(t)$ at the cavity wall and p_∞ far away

Location	Condition	Origin
$r = R_c(t)$	$v_r = \dot{R}_c$	the liquid at the cavity wall moves with it
$r \to \infty$	$v_r \to 0, p \to p_\infty$	liquid at rest far away
$r = R_c(t)$	$\bar{\bar{\tau}} \cdot \bar{n} = 0, \bar{n} = \bar{e}_r$	cavity pressure neglected, no traction
$t = 0$	$R_c = R_0, \dot{R}_c = V_0$	statement

PHASE 3

Step-by-Step Derivation

3.1 Velocity and irrotationality (item 1)

$$\frac{1}{r^2}\frac{\partial(r^2 v_r)}{\partial r} = 0 \implies r^2 v_r = A(t), \qquad v_r(R_c,t) = \dot{R}_c \implies A(t) = R_c^2 \dot{R}_c \tag{1}$$

$$\boxed{v_r(r,t) = \frac{R_c^2 \dot{R}_c}{r^2}} \tag{2}$$

The vorticity of $\bar{v} = v_r(r,t)\bar{e}_r$, from the curl in spherical coordinates, has the components $\omega_r = \frac{1}{r\sin\theta}\left[\frac{\partial(v_\phi \sin\theta)}{\partial\theta} - \frac{\partial v_\theta}{\partial\phi}\right] = 0, \omega_\theta = \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial v_r}{\partial\phi} - \frac{\partial(r v_\phi)}{\partial r}\right] = 0$ and $\omega_\phi = \frac{1}{r}\left[\frac{\partial(r v_\theta)}{\partial r} - \frac{\partial v_r}{\partial\theta}\right] = 0$, because $v_\theta = v_\phi = 0$ and v_r does not depend on θ or ϕ. Hence $\nabla \wedge \bar{v} = 0$: **the flow is irrotational**, and therefore $\nabla \wedge (\nabla \wedge \bar{v}) = 0$.

3.2 Pressure field (item 2)

The viscous force of the incompressible flow is $-\mu\nabla \wedge (\nabla \wedge \bar{v}) = 0$, so the r component of (4.41) is the Euler equation:

$$\rho\left(\frac{\partial v_r}{\partial t} + v_r\frac{\partial v_r}{\partial r}\right) = -\frac{\partial p}{\partial r} \tag{3}$$

With $\partial v_r/\partial t = (R_c^2\ddot{R}_c + 2R_c\dot{R}_c^2)/r^2$ (product rule on $R_c^2\dot{R}_c$) and $v_r\,\partial v_r/\partial r = -2R_c^4\dot{R}_c^2/r^5$:

$$\frac{\partial p}{\partial r} = -\rho\left[\frac{R_c^2\ddot{R}_c + 2R_c\dot{R}_c^2}{r^2} - \frac{2R_c^4\dot{R}_c^2}{r^5}\right] \tag{4}$$

Integrating from r to ∞ with $p(\infty) = p_\infty$ (the antiderivatives are $-1/r$ and $-1/(4r^4)$, and Barrow's rule gives $1/r$ and $1/(4r^4)$):

$$\boxed{\frac{p(r,t) - p_\infty}{\rho} = \frac{R_c^2 \ddot{R}_c + 2R_c \dot{R}_c^2}{r} - \frac{R_c^4 \dot{R}_c^2}{2r^4}} \tag{5}$$

3.3 Equation at the cavity wall (item 3)

At $r = R_c(t)$, (5) gives $\dfrac{p_R - p_\infty}{\rho} = R_c \ddot{R}_c + 2\dot{R}_c^2 - \dfrac{\dot{R}_c^2}{2}$:

$$\boxed{R_c \ddot{R}_c + \frac{3}{2}\dot{R}_c^2 = \frac{p_R(t) - p_\infty}{\rho}, \qquad R_c(0) = R_0, \;\; \dot{R}_c(0) = V_0} \tag{6}$$

3.4 Pressure at the wall: null stress (item 4)

The unit normal to the cavity wall, pointing into the liquid, is $\bar{n} = \bar{e}_r$. The stress vector is $\bar{\bar{\tau}} \cdot \bar{e}_r = (-p + \tau'_{rr})\bar{e}_r + \tau'_{r\theta}\bar{e}_\theta + \tau'_{r\phi}\bar{e}_\phi$. For this flow $\tau'_{r\theta} = \tau'_{r\phi} = 0$ and (Notes.pdf, stress components in spherical coordinates)

$$\tau'_{rr} = 2\mu \frac{\partial v_r}{\partial r} = -\frac{4\mu R_c^2 \dot{R}_c}{r^3}, \qquad \tau'_{rr}\Big|_{R_c} = -\frac{4\mu \dot{R}_c}{R_c} \tag{7}$$

The null-stress condition at $r = R_c$ gives $-p_R + \tau'_{rr}(R_c) = 0$:

$$\boxed{p_R(t) = -\frac{4\mu \dot{R}_c}{R_c}} \tag{8}$$

3.5 Closed ODE and non-dimensionalization (item 5)

Substituting (8) in (6) with $\nu = \mu/\rho$:

$$R_c \ddot{R}_c + \frac{3}{2}\dot{R}_c^2 + \frac{4\nu \dot{R}_c}{R_c} = -\frac{p_\infty}{\rho} \tag{9}$$

With $R_c = R_0 R$ and $t = R_0 \tau / V_0$: $\dot{R}_c = V_0\, R'$ and $\ddot{R}_c = (V_0^2/R_0) R''$ (chain rule, $d/dt = (V_0/R_0)\, d/d\tau$, with $R' = dR/d\tau$). Substituting and dividing by V_0^2:

$$\boxed{R\,R'' + \frac{3}{2}R'^2 + \frac{4}{\mathrm{Re}}\frac{R'}{R} = -\Lambda, \qquad R(0) = 1,\;\; R'(0) = 1} \tag{10}$$

with $\mathrm{Re} = \rho V_0 R_0/\mu$ (because $\nu/(V_0 R_0) = 1/\mathrm{Re}$) and $\Lambda = p_\infty/(\rho V_0^2)$. The problem depends only on these two parameters.

3.6 Limit $\mathrm{Re} \to \infty$ (item 6)

The viscous term drops out: $R\,R'' + \frac{3}{2}R'^2 = -\Lambda$. With $z = \ln R$ and $F = (dR/d\tau)^2$, the chain rule gives $dF/d\tau = 2R'R''$ and $dz/d\tau = R'/R$, so

$$\frac{dF}{dz} = \frac{dF/d\tau}{dz/d\tau} = 2R\,R'' \tag{11}$$

and the equation becomes $\frac{1}{2}\frac{dF}{dz} + \frac{3}{2}F = -\Lambda$, that is $\frac{dF}{dz} + 3F = -2\Lambda$ with $F(0) = R'(0)^2 = 1$ (at $\tau = 0$, $R = 1$ and $z = 0$). Multiplying by the integrating factor e^{3z}: $(F\,e^{3z})' = -2\Lambda e^{3z}$, so $F\,e^{3z} = -\frac{2\Lambda}{3}e^{3z} + C$ and $F(0) = 1$ gives $C = 1 + 2\Lambda/3$:

$$F(z) = e^{-3z} + \frac{2\Lambda}{3}\left(e^{-3z} - 1\right) \tag{12}$$

With $e^{-3z} = R^{-3}$:

$$\boxed{\frac{dR}{d\tau} = \left[R^{-3} + \frac{2\Lambda}{3}\left(R^{-3} - 1\right)\right]^{1/2}} \tag{13}$$

(positive root while the cavity expands).

3.7 Maximum radius and time to reach it (item 7)

The cavity stops growing when $dR/d\tau = 0$: $R_{\max}^{-3}(1 + 2\Lambda/3) = 2\Lambda/3$, that is

$$\boxed{R_{\max} = \left(1 + \frac{3}{2\Lambda}\right)^{1/3}} \tag{14}$$

Separating variables in (13), $d\tau = dR/\sqrt{R^{-3} + \frac{2\Lambda}{3}(R^{-3} - 1)}$, and integrating from $R = 1$ at $\tau = 0$ to $R_{\max}$ at $\tau_{\max}$ (with the dummy variable x):

$$\boxed{\tau_{\max}(\Lambda) = \int_1^{(1+3/(2\Lambda))^{1/3}} \frac{dx}{\sqrt{x^{-3} + \frac{2\Lambda}{3}\left(x^{-3} - 1\right)}}} \tag{15}$$

Approximation for $\Lambda \ll 1$. Then $R_{\max} \simeq (3/(2\Lambda))^{1/3} \gg 1$. Factor x^{-3} out of the square root: the integrand is $x^{3/2}/\sqrt{(1 + 2\Lambda/3) - (2\Lambda/3)x^3}$ and, with $2\Lambda/3 \simeq R_{\max}^{-3}$ for small Λ,

$$\tau_{\max} \simeq \int_1^{R_{\max}} \frac{x^{3/2}\,dx}{\sqrt{1 - (x/R_{\max})^3}} \simeq R_{\max}^{5/2} \int_0^1 \frac{s^{3/2}\,ds}{\sqrt{1 - s^3}} \tag{16}$$

(with $x = R_{\max} s$, and the lower limit $s = 1/R_{\max} \to 0$). The substitution $u = s^3$ gives $s^{3/2} ds = \tfrac{1}{3} u^{-1/6} du$, so the integral is $\tfrac{1}{3} B(\tfrac{5}{6}, \tfrac{1}{2}) = \dfrac{\sqrt{\pi}\,\Gamma(5/6)}{3\,\Gamma(4/3)} \simeq 0.747$:

$$\boxed{\tau_{\max} \simeq 0.747 \left(\frac{3}{2\Lambda}\right)^{5/6} = 0.747 \left(\frac{3}{2}\right)^{5/6} \Lambda^{-5/6}} \qquad (\Lambda \ll 1) \tag{17}$$

Discrepancy with the official solution

The handwritten solution approximates the integrand by $x^{3/2}$ over the whole range and obtains $\tau_{\max} \simeq \frac{2}{5}(3/2)^{5/6}\Lambda^{-5/6}$, which neglects the factor $(1 - x^3/R_{\max}^3)^{-1/2}$ of (16). Integrating (15) numerically gives $\tau_{\max} = 48.3,\ 330.8,\ 2255.4$ for $\Lambda = 10^{-2},\ 10^{-3},\ 10^{-4}$, while (17) gives $48.6,\ 331.1,\ 2255.8$ and the sheet's expression $26.0,\ 177.3,\ 1208.2$. The scaling $\Lambda^{-5/6}$ is the same; the computed coefficient 0.747 is kept.

3.8 Limit $\mathrm{Re} \to 0$ (item 8)

For $\mathrm{Re} \to 0$ the viscous term dominates the inertial terms and (10) reduces to the first-order ODE

$$\frac{4}{\mathrm{Re}}\frac{R'}{R} \simeq -\Lambda \implies \frac{dR}{d\tau} = -\frac{\Lambda\,\mathrm{Re}}{4}\,R \tag{18}$$

whose solution with $R(0) = 1$ is

$$R(\tau) \simeq \exp\left(-\frac{\Lambda\,\mathrm{Re}}{4}\,\tau\right) \tag{19}$$

It is not possible to satisfy both initial conditions: a first-order ODE admits only one, and (19) has $R'(0) = -\Lambda\mathrm{Re}/4 \neq 1$. The approximation is valid only after an initial transient, of duration of the order of Re in τ, during which the inertial terms $R\,R'' + \frac{3}{2}R'^2$ cannot be neglected and the initial velocity is dissipated by viscosity. In this limit the ambient pressure p_∞ crushes the cavity, which then collapses exponentially instead of expanding.

PHASE 4

Interpretation, Limits and Dimensional Check

- **Physical meaning.** (6) is the Rayleigh equation for the cavity with the viscous correction (8): the cavity decelerates because of the ambient pressure p_∞ (term Λ) and the viscous normal stress (term $4/(\mathrm{Re}\,R)$). For $\mathrm{Re}\to\infty$ the cavity stops at $R_{\max}$ and then collapses; for $\mathrm{Re}\to 0$ viscosity dominates.
- **Limits.** For $\Lambda\to 0$ the right-hand side of (13) is $R^{-3/2}$ with no stopping point: the cavity expands for ever in the inviscid, pressureless case. For $\Lambda\to\infty$, $R_{\max}\to 1$: the cavity cannot expand.
- **Dimensional check.** $R_c\ddot{R}_c$ and $\dot{R}_c^2$ have units $\mathrm{m}^2/\mathrm{s}^2$, as does p_∞/ρ (Pa/(kg/m^3)). $4\nu\dot{R}_c/R_c$ has units $(\mathrm{m}^2/\mathrm{s})(\mathrm{m}/\mathrm{s})/\mathrm{m}=\mathrm{m}^2/\mathrm{s}^2$. Re and Λ are dimensionless. $p_R=-4\mu\dot{R}_c/R_c$ has units Pa.
- **Consistency.** At $R=1$ the bracket of (13) equals 1 for every Λ, so $R'(0)=1$: the solution keeps the initial velocity V_0.